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MILITARY SPECIFICATION

FILTER UNIT, FLUID, PRESSURE MXU-408/M, ABSOLUTE 5 MICRON, HYDRAULIC

1. SCOPE

1.1 This specification covers a hydraulic fluid filter unit with a replaceable element for retaining particles larger than 5 microns and 96.6 percent by weight of 5 micron and smaller particles.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-831	Test Reports, Preparation of
MS3102	Connector, Receptacle, Electric, Box Mounting

FSC 4920

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MS33586	Metals, Definition of Dissimilar
MS33649	Bosses, Fluid Connection - Internal Straight Thread
MS33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal

Air Force-Navy Aeronautical

AN814	Plug and Bleeder - Screw Thread

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. Filter units and filter elements furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.2 Components. The filter unit shall consist of the following major components:

Description	See Requirement
Housing	3.7.1
Element	3.7.2
Differential pressure indicator	3.7.3.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Protective treatment. When materials are used in the construction of the filter unit that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with use, age, or extremes of climatic and environmental conditions shall be avoided.

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3.4.2 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to water, hydraulic fluids, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.4.3 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.5 Design and construction. The filter unit shall be so designed and constructed that no parts will work loose in service. The filter unit shall withstand the stresses, jars, vibrations, and other conditions incident to shipping, storage, installation, and flight line service.

3.5.1 Functional design. The filter unit shall be designed as a lightweight filter and for use with hydraulic fluids conforming to MIL-H-5606 with flows to 30 gpm and operating pressures to 3,000 psi.

3.5.2 Reliability. The filter unit shall be designed and constructed as specified herein and to successfully complete the tests of 4.4.3 with no failures preventing satisfactory performance.

3.5.3 Maintainability. The filter unit shall be designed and constructed as specified herein and to provide the following:

a. Minimum number of parts consistent with the reliability and performance specified herein.

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing.

c. Permit maintenance with general-purpose tools and equipment normally available commercially. Use of special-purpose tools and equipment shall be subject to procuring activity approval.

d. Permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other equipment parts or components (see 3.7.1.1). Parts and components shall be located for ample and rapid access, unless performance or reliability will be appreciably degraded by the accessible location.

e. Minimum number of tools required for maintenance by design practices such as reducing the number of different bolthead sizes to the practicable minimum.

3.5.4 Foolproofness. Where improper installation of an item could cause malfunctioning of that item or the system in which it is installed, an unsymmetrical mounting means shall be provided. This mounting shall be so designed that the item can only be installed in its proper operating position.

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3.5.4.1 Element installation. The filter unit shall be so designed that it will be impossible to assemble the filter unit, if the filter element is incorrectly installed.

3.6 Performance. The filter unit shall perform satisfactorily when subjected to the following:

a. Operating pressures up to 3,000 psig and proof pressures up to 7,500 psig

b. With a clean element installed, the differential pressure drop shall not exceed 50 psi between the inlet and outlet of the housing at 30 gpm flow and a fluid temperature of 100°F

c. Actuation of the differential pressure indicator at a differential pressure of 100 ±15 psi between the inlet and outlet ports of the housing

d. Operating temperatures from -65° to +160°F

e. Burst pressures up to 11,250 psi.

3.6.1 Vibration. The filter unit shall be capable of withstanding the vibrations specified in 4.6.6.5.1.

3.7 Details of components

3.7.1 Housing. The housing shall be so designed that flow through the element will be from the outside to the inside. Flow entering the housing shall not impinge directly upon the element filter media.

3.7.1.1 Housing design. The design of the housing shall permit change of the element without removal of the inlet or outlet line.

3.7.1.2 Pressure indicator. Provisions shall be incorporated in the housing for the differential pressure indicator specified in 3.7.3.

3.7.1.3 Dimensions. The inside dimensions and the inlet and outlet threads of the housing shall be as shown on figure 1.

3.7.1.4 Mounting. Provisions shall be incorporated on the housing, or a corrosion-resistant adapter shall be provided, to permit its being mounted on a flat vertical surface with four 11/32-inch diameter holes located in a rectangular position with hole centers spaced 4-1/2 ±1/64 by 10 ±1/64 inch. The long axis of the rectangle shall be parallel to the axis of the filter elements. The top pair of holes shall be 3-3/8 ±3/16 inch below the port-to-port centerline. The port-to-port centerline shall be parallel to the flat vertical mounting surface. In this position, with the port's end of the filter upward, the outlet port shall be on the right side. The distance from the port-to-port centerline to the vertical mounting surface shall be 3-1/2 ±1/8 inch.

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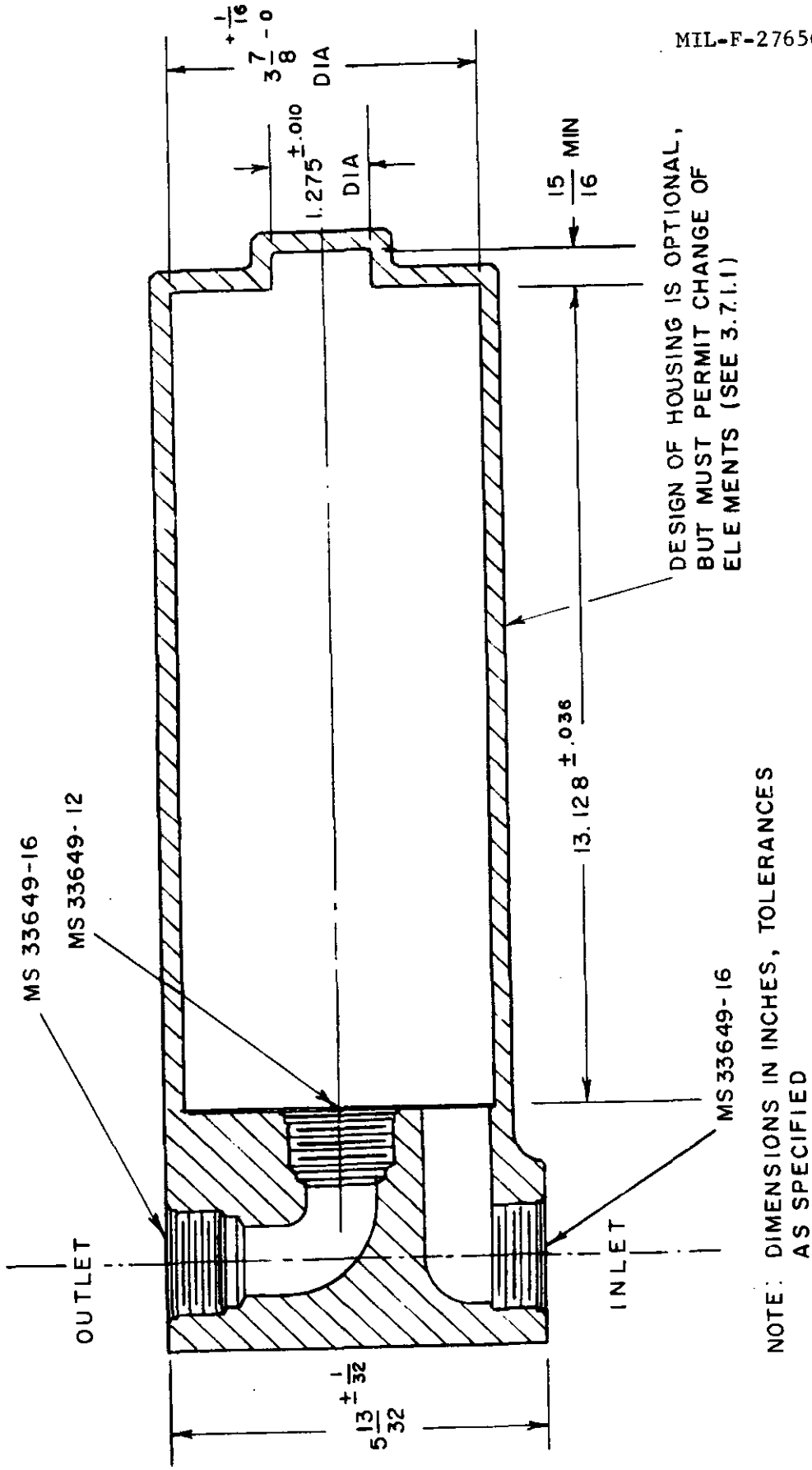


FIGURE 1. Dimensions of Housing

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3.7.1.5 Wrench pad and drain port. The bottom of the housing shall have a 2-1/4-inch hex wrench pad and shall incorporate an MS33649-4 drain port with an AN814-4 plug.

3.7.1.6 Bleed port. The top of the housing shall incorporate an MS33649-4 air bleed port, located such that optimum bleeding of entrapped air within the filter will be insured. The bleed port shall be fitted with an AN814-4 plug.

3.7.2 Element

3.7.2.1 Media. The filtering media shall be included in a self-contained cleanable or throwaway element with dimension envelope conforming to figure 2. The element may consist of one or two parts, with the parts being either of a cleanable or throwaway type in any combination. Applicable packing shall be furnished with the element for its installation in the housing.

3.7.2.2 Particle size. The element shall remove 96.6 percent by weight of all particles 5 microns and smaller in size, and shall remove 100 percent of all particles over 5 microns from fluid at pressures of 0 to 3,000 psi and at temperatures of +50° to +160°F.

3.7.2.3 Element material. The element shall be of a material that will not change the composition of, or be affected by, hydraulic fluids conforming to MIL-H-5606.

3.7.2.4 Element performance. The element shall be of a material to maintain filtering quality and shall be supported to withstand differential pressures as follows:

a. The maximum particle size passed while withstanding 0 to 150 psi differential pressure shall be 5 microns.

b. The maximum particle size passed while withstanding 150 to 5,000 psi differential pressure shall be 18 microns.

3.7.2.5 Element life. Cleanable elements or parts shall withstand 1,000 hours of operation at 160°F. Noncleanable (throwaway) elements or parts shall withstand 100 hours of operation at 160°F.

3.7.2.6 Cleanable element materials. Cleanable elements shall be of materials which will not be adversely affected by the cleaning procedures specified in 4.6.6.4f.

3.7.2.7 Dirt-holding capacity. The element shall retain a minimum of 40 grams of AC fine test dust or 200 grams of carbonyl iron E with a differential pressure not exceeding 115 psi, a fluid temperature of 100°F, and a flow of 30 gpm.

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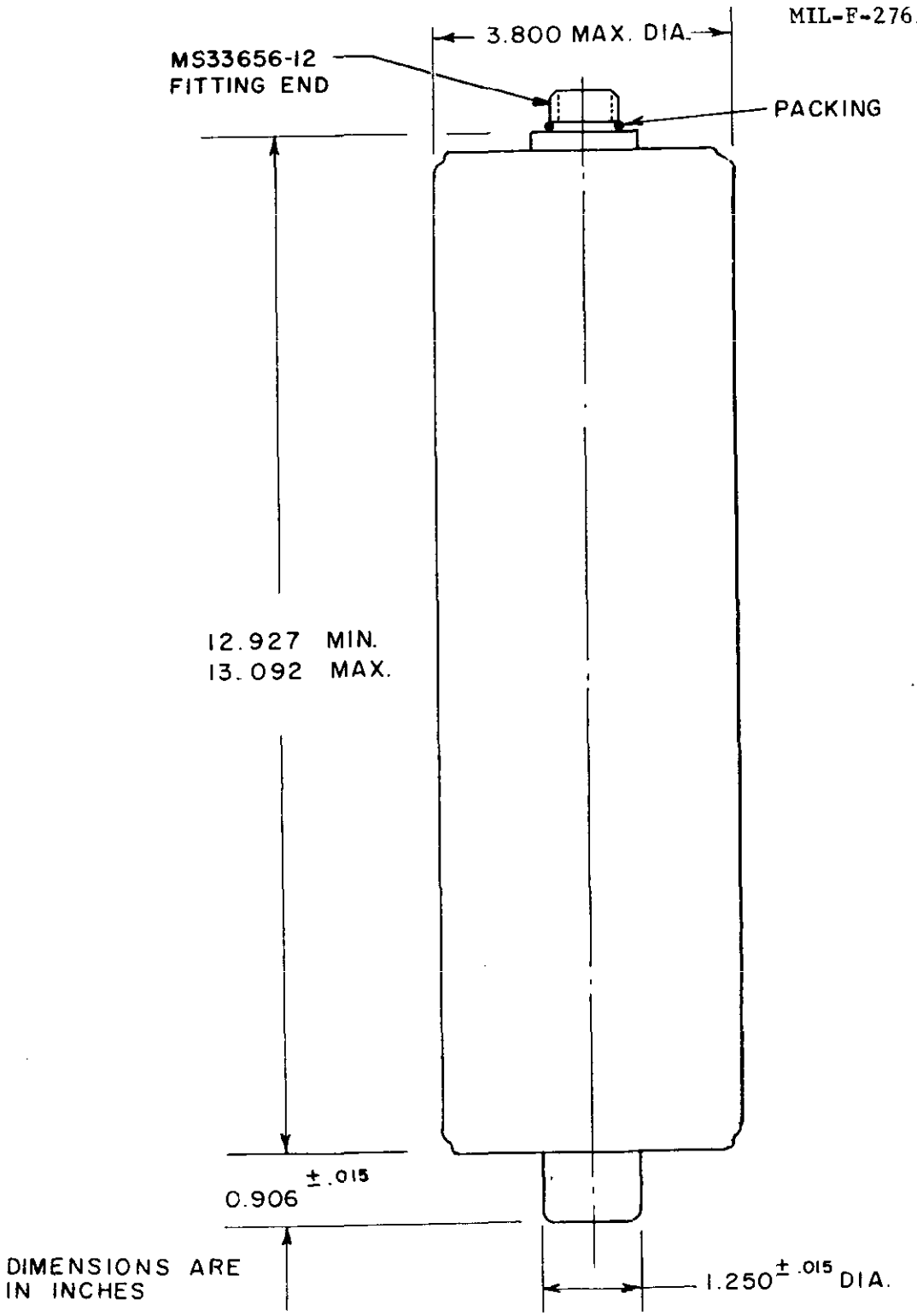


FIGURE 2. Element Envelope Dimensions

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3.7.2.8 Handling damage. If applicable, the element shall be provided with a perforated cover or other device to minimize element handling damage.

3.7.3 Differential pressure indicator. The filter unit shall incorporate an integral device capable of actuating a 24-V, d-c or 110-V, a-c electrical signal when the differential pressure across the unit exceeds 100 ±15 psi. Once actuated, the signal shall remain energized until the device is manually reset. The manual reset device shall be external and easily accessible. The indicator shall incorporate an MS3102R10SL-3P electrical receptacle with pin arrangement as follows:

Pin A	-	Normally open
Pin B	-	Common
Pin C	-	Normally closed.

3.8 Interchangeable components. All filter units and filter elements listed on the Qualified Products List shall be functionally and dimensionally interchangeable.

3.8.1 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.9 Weight. The dry weight of the filter unit shall not exceed 55 pounds.

3.10 Operational markings

3.10.1 Direction of flow. The direction of flow shall be clearly and permanently indicated by at least two arrows marked on opposite sides of the housing or ports, or the ports shall be clearly marked INLET and OUTLET.

3.10.2 Instruction placard. Each filter unit shall be provided with a permanently mounted instruction placard. The placard shall describe installation and maintenance procedures and shall show an illustrated parts breakdown.

3.11 Identification of product. The filter unit shall be marked for identification in accordance with MIL-STD-130 and as follows:

3.11.1 Manufacturer's part number. The manufacturer's part number and drawing number shall be the same.

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3.11.2 Housing nameplate. The housing nameplate shall be permanently marked with the following information:

Fluid specification: MIL-H-5606

Maximum flow: 30 gpm

Maximum pressure: 5,000 psi

Procurement data:

3.11.3 Replaceable element. The replaceable element shall be permanently marked with the following information. If marked on a sealing surface, the markings shall not affect the sealing of the element.

Manufacturer's part number:

Manufacturer's name or trademark:

Cleanable or noncleanable (as applicable):

Element rating: 5 micron absolute

Fluid specification: MIL-H-5606

Procurement data:

3.12 Workmanship. The filter unit, including all parts and assemblies, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of tests. The inspection and testing of the filter unit shall be classified as follows:

a. Qualification tests . . . . . See 4.4

b. Acceptance tests. . . . . See 4.5.

4.3 Test conditions

4.3.1 Test fluid. The hydraulic fluid used for all tests shall conform to MIL-H-5606.

4.3.2 Fluid temperature. Unless otherwise specified herein, the tests shall be conducted with the fluid at 100°F. The oil temperature during each test specified herein shall be noted in the test report. Unless otherwise specified herein, all temperatures shall be held within ±10°F.

4.3.3 Ambient conditions. Unless otherwise specified herein, tests shall be conducted at standard atmospheric conditions (approximately 68°F and 29.92 inches Hg).

4.4 Qualification tests

4.4.1 Test samples. One filter unit and five elements, or five sets of element parts, shall be subjected to the qualification tests specified in 4.4.3.

4.4.1.1 Element qualification. It is permissible to qualify the element only. In this case, five sets of elements shall be subjected to tests as specified in 4.4.3 with omission of the 4.6.2 and 4.6.4 tests.

4.4.2 Test report. After completion of the qualification tests, a test report shall be prepared in accordance with MIL-STD-831 and three complete copies of the report furnished to the procuring activity.

4.4.2.1 Reliability and maintainability information. The following information shall be submitted as an attachment accompanying the qualification test report (same quantity and quality of copies) or shall be included as part of that report:

a. All failures, maintenance, and other events recorded shall be identified by accumulated operating time, miles, cycles, or position in the test procedure as appropriate. Test conditions during failures or irregular operations shall be recorded.

b. Description of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations.

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c. Description of the engineering reasoning behind any corrections made, to be made on production items, or proposed to be made and behind the predicted effectiveness of those corrections.

d. Test activity or contractor comments on field conditions or procedures to be avoided or cultivated to increase the responsibility and useful life of the item.

e. Estimates (±5 percent of actual experience) of manhours required for each maintenance and servicing action during the tests. A brief description of the qualifications and experience of the personnel involved shall be included and shall be adequate to permit comparison to the personnel anticipated in similar field work.

f. Summary of the above information and of the results and contractor or test activity recommendations for modifications of equipment requirements, tests, or features that should improve future equipment.

4.4.3 Qualification tests. The qualification tests for the filter unit shall consist of all the tests described under 4.6. The qualification test for the element alone shall consist of the same tests described under 4.6 with the omission of 4.6.2 and 4.6.4. The filter housing and differential pressure indicator utilized in these element qualification tests shall be components of a filter unit which has met the qualification tests. All qualification tests shall be conducted in the order listed below.

Filter unit

a.	Examination of product	4.6.1
b.	Proof pressure test	4.6.2
c.	Pressure drop test	4.6.3
d.	Differential pressure indicator operation test	4.6.4
e.	Extreme temperature operation test	4.6.5
f.	Burst pressure test	4.6.9

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Elements

Element No. 1		
a.	Examination of product	4.6.1
b.	Bubble test	4.6.6.1
c.	Degree of filtration test	4.6.6.2
d.	Pressure impulse tests	4.6.7
e.	Bubble test	4.6.6.1
f.	Flow fatigue tests	4.6.8
g.	Bubble test	4.6.6.1
Element No. 2		
a.	Examination of product	4.6.1
b.	Bubble test	4.6.6.1
c.	Pressure buildup and collapse (test dust)	4.6.6.4
d.	Bubble test	4.6.6.1
Element No. 3		
a.	Examination of product	4.6.1
b.	Bubble test	4.6.6.1
c.	Largest particle passed test	4.6.6.3
d.	Bubble test	4.6.6.1
e.	Pressure buildup and collapse (iron E)	4.6.6.4
f.	Bubble test	4.6.6.1

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Element No. 4

a.	Examination of product	4.6.1
b.	Bubble test	4.6.6.1
c.	Media migration and vibration tests	4.6.6.5
d.	Bubble test	4.6.6.1

Element No. 5

a.	Bubble test	4.6.6.1
b.	Fluid composition test	4.6.10
c.	Bubble test	4.6.6.1

4.5 Acceptance tests

4.5.1 Housing tests. Each housing shall be subjected to the following tests:

a. Examination of product . . . . . See 4.6.1

b. Proof pressure test. . . . . See 4.6.2

c. Two cycles of the differential pressure indicator operation test . . . . . See 4.6.4.

4.5.2 Element tests. Each element shall be subjected to the following tests:

a. Examination of product . . . . . See 4.6.1

b. Bubble test . . . . . See 4.6.6.1.

4.6 Test methods

4.6.1 Examination of product. The filter unit shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and marking.

4.6.2 Proof pressure test. The filter unit shall be assembled with the outlet port plugged. The filter unit shall then be filled with clean test fluid and a pressure of 7,500 psig applied at least twice and held for 2 minutes at each application. The pressure shall be reduced to zero between applications. Any evidence of permanent deformation, malfunction, or leakage shall be cause for rejection.

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4.6.3 Pressure drop test. The plug in the outlet end of the filter unit shall be removed and a pressure indicator installed at the outlet and inlet of the housing. The indicators shall be accurate to within ±1/2 percent of full-scale reading. Elements which have successfully withstood the tests specified in 4.6.1 and 4.6.6.1 shall be installed and a flow of 30 gpm with the oil temperature stabilized at 100°F shall be directed through the unit. The differential pressure from the inlet to the outlet shall not exceed 50 psi.

4.6.4 Differential pressure indicator operation test. With the outlet of the filter unit partially blocked, the pressure at the inlet to the filter unit shall be slowly increased until the differential pressure indicator is fully actuated. For this test, a 24- or 110-V power supply and test light shall be wired through the pressure indicator device. The test light shall illuminate when the differential pressure between the inlet and outlet reaches 100 ±15 psi. The pressure shall be raised to 5,000 psi (the outlet may be totally blocked for this portion of the test). The pressure shall be returned to zero, and the test light shall remain illuminated. The indicator shall be manually reset. The above shall be considered one complete cycle and shall be conducted 24 times. Twelve cycles shall be accomplished using 24-V input, and twelve cycles shall be accomplished using 110-V input. Failure of the indicator to actuate at 100 ±15 psi on any one of the above 24 cycles shall be cause for rejection.

4.6.5 Extreme temperature operation test. The filter unit shall be subjected to -65°F for 24 hours. During this period, pressure applied shall not exceed 10 psi. At the end of this period, a differential pressure of 100 ±15 psi shall be created by applying flow through the unit and continuing this flow for 10 minutes. The differential pressure indicator shall actuate within its specified range. The flow during this test shall be recorded. Proof pressure shall then be applied to the inlet port for 2 minutes with the outlet blocked. The fluid used for these tests shall be not warmer than -65°F. This test shall then be repeated at -40° and +20°F. The unit shall then be removed from the cold chamber and placed in a temperature environment of +160°F within 20 minutes. Flow shall be resumed until a differential pressure of 100 ±15 psi has been obtained. The differential pressure indicator shall actuate within its specified range. The element shall be removed and shall successfully withstand the test specified in 4.6.6.1.

4.6.6 Element tests

4.6.6.1 Bubble test. The test element shall be installed in a setup similar to that shown on figure 3. Should the design of the filter unit make it impractical to conduct the bubble test with airflow from inside-out, the test may be conducted with airflow from outside-in provided the test setup is approved by the procuring activity. The fluid level shall be maintained 1/2 inch above the test element. The element shall be slowly rotated while the air pressure

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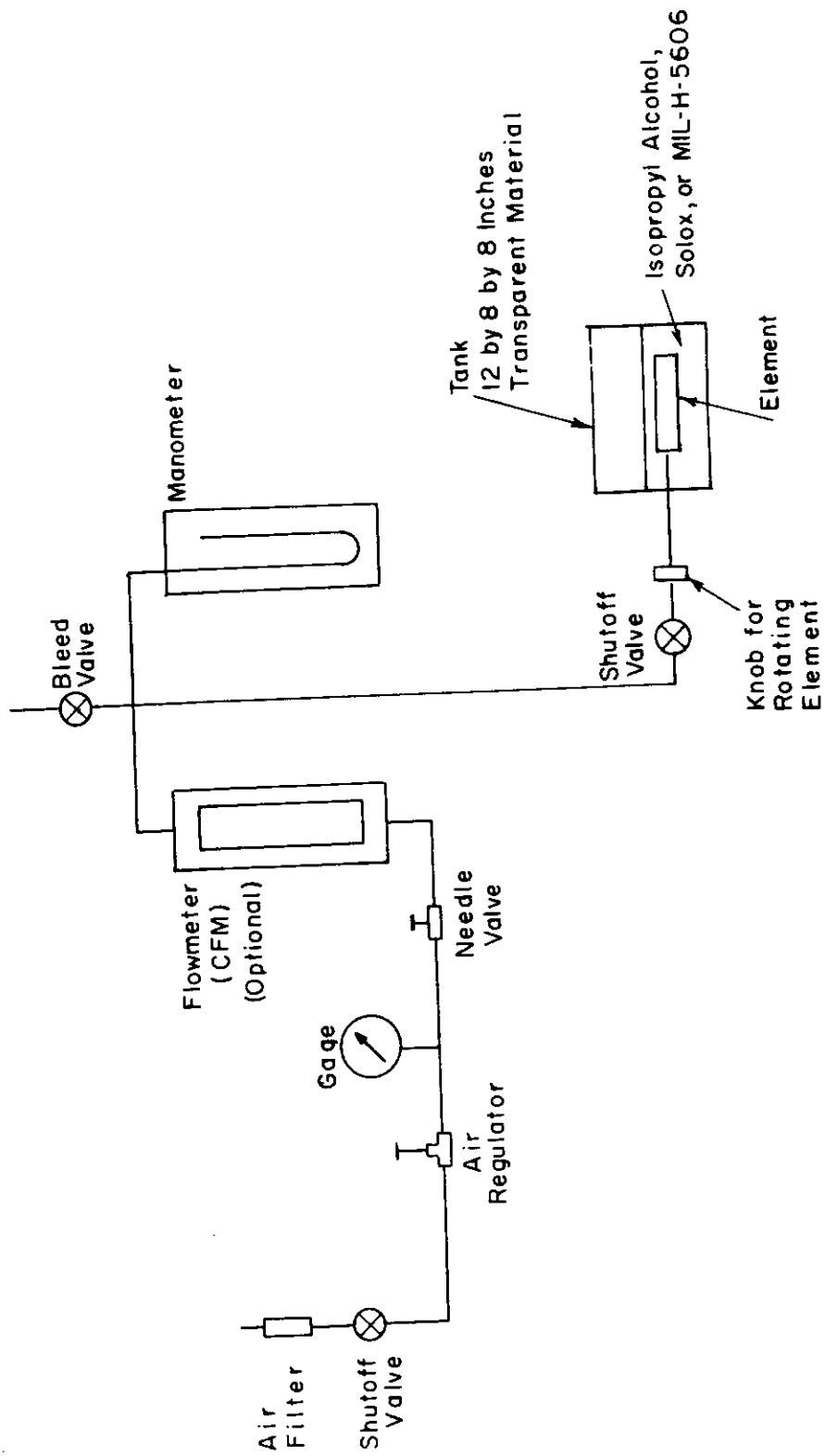


FIGURE 3. Typical Schematic Diagram for Air Bubble Test

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is gradually increased by means of the needle valve. The area of greatest porosity shall be determined by observing the first bubble on the surface of the element and the manometer reading in inches of water at which this bubble emerges from the element. This shall be recorded as the initial bubble point. The initial bubble-point pressure for each element shall be not less than 0.90 times the minimum qualification bubble point.

4.6.6.2 Degree of filtration test. The hydraulic fluid and air used in the degree of filtration test shall be clean and filtered prior to the test. Figure 4 shows a degree of filtration test setup with a cleanup device attached.

4.6.6.2.1 Test procedure for determination of oil cleanliness. The test procedure for determination of oil cleanliness shall be as follows:

a. A cleanup apparatus consisting of needle valves, a Millipore fixture, Millipore Type HA disc (1-inch diameter) manometer, flowmeter, and pump with adjustable internal bypass shall be installed on the test stand similar to the setup shown on figure 4.

b. A constant flow shall be established and maintained for 30 minutes, with at least periodic agitation of the contaminant mixing chamber, at a pressure differential of 50 to 60 psi across the Millipore filter assembly.

c. With the flow held constant, the pressure drop across the Millipore fixture shall be recorded at the beginning and end of the flow period.

d. The pressure drop recorded shall not exceed 1-1/2 percent of the initial value. If the variation exceeds 1-1/2 percent, the system shall be recleaned until the 1-1/2 percent maximum variation is achieved.

e. Two thousand milliliters of the clean fluid shall be collected in a 5,000-ml beaker from the sampling valve downstream of the test filter housing.

f. A membrane filter 47-mm diameter disk, absolute 1.0 micron (type AA Millipore, or equal), shall be washed with 200 ml of petroleum ether and dried to constant weight at 125°F. The weight shall be accurate to 0.1 mg. The disk shall be heated at 125°F for 30 minutes, then cooled 30 minutes in a desiccator.

g. The weighed filter membrane shall be assembled into the filter holder assembly (see figure 5) and fitted to a 5,000-ml vacuum flask connected to a vacuum pump (minimum 25 inches Hg vacuum).

h. The 2,000-ml fluid sample shall be filtered through the membrane disk. The beaker shall be washed with 800 ml of petroleum naptha and 800 ml of petroleum ether (boiling point 95° to 130°F). The wash fluid shall be filtered through the membrane disk. One hundred milliliters of petroleum naptha

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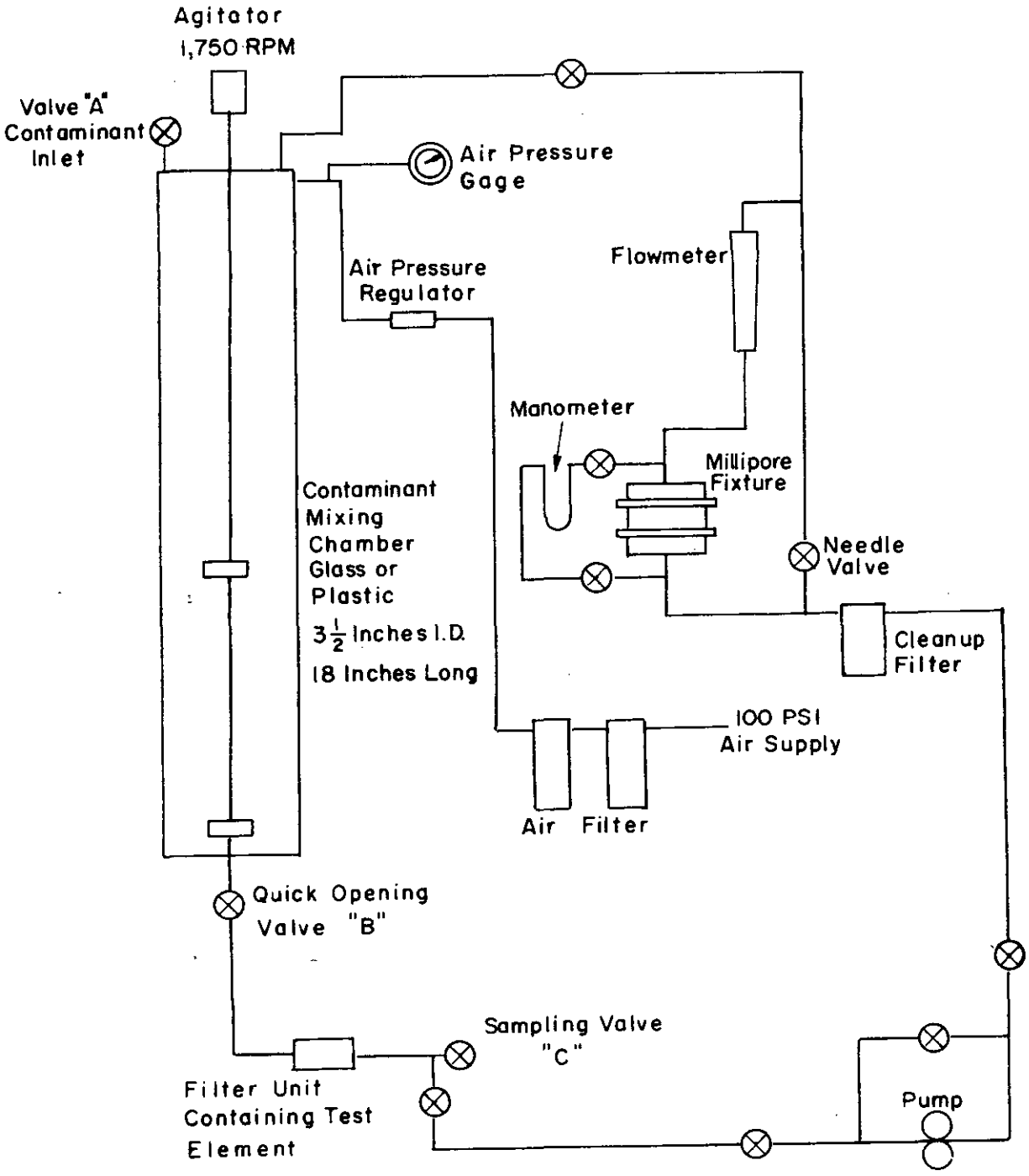


FIGURE 4. Typical Apparatus for Determination of Efficiency of Filter Element

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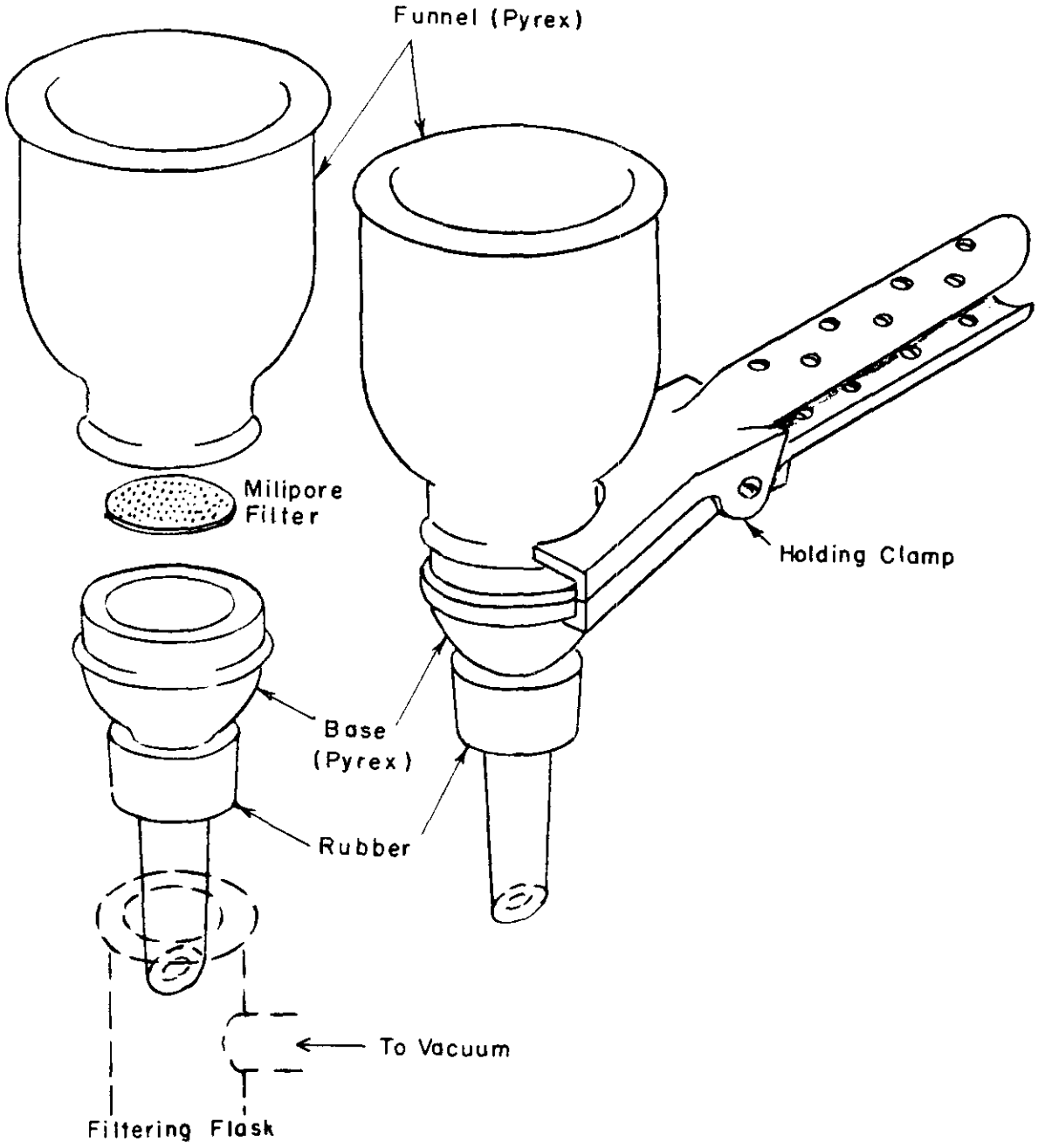


FIGURE 5. Diagram of Effluent Filtration Setup

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and one hundred milliliters of petroleum ether shall then be used to wash down the funnel, and then this wash shall be filtered.

i. The membrane disk shall then be removed and dried to a constant weight at 125°F and weighed to an accuracy of 0.1 mg. The contaminant weight shall be less than 0.0004 gram/2,000 ml.

4.6.6.2.2 Filter element tests. The degree of filtration for filter elements shall be determined as follows:

a. The system shall be checked for oil cleanliness in accordance with 4.6.6.2.1.

b. Prefiltered hydraulic fluid shall be added through plug valve A to bring the level in the mixing chamber to 2,000 ml. The element shall be installed in the filter housing.

c. Valve B shall be closed.

d. A 10-ml slurry containing 3 grams of contaminant shall be added to the fluid through a small funnel inserted in plug valve A. The contaminant shall be 1.5 grams of APM F-9 beads (see figure 6) and 1.5 grams of standardized fine air cleaner (AC)) test dust (see 4.6.6.4d).

e. The contaminant shall be uniformly distributed by churning the fluid for 5 minutes with an agitator.

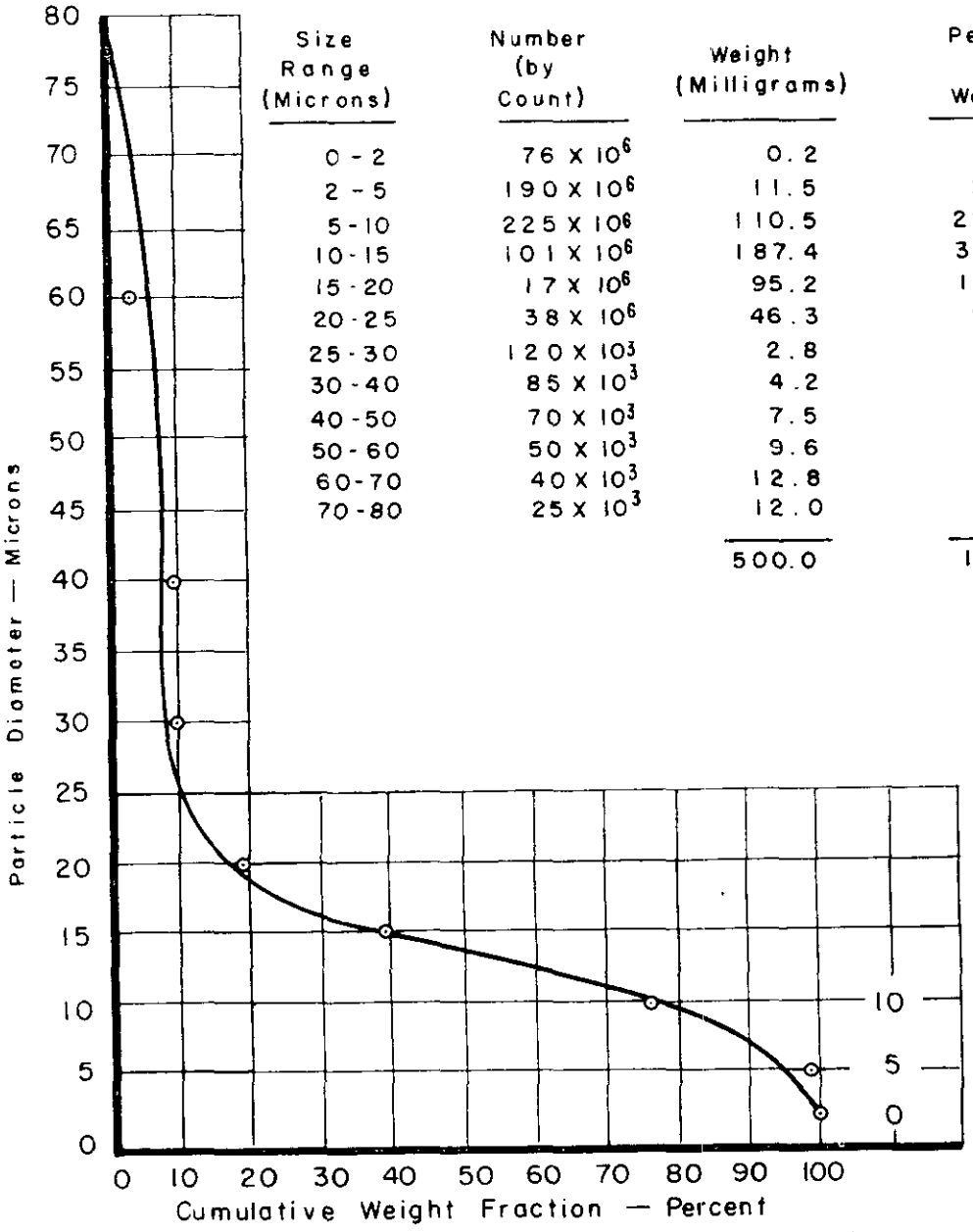
f. Plug valve A shall be closed, and the glass chamber containing hydraulic fluid and contaminant shall be pressurized by the air regulator. The air regulator shall be used to maintain flow. Sufficient air pressure shall be employed to maintain rated flow.

g. Gate valve B shall be opened and air pressure shall force fluid containing contaminant through the sample filter unit at rated flow. This 2,000 ml of filtrate shall be collected in a 5,000-ml beaker from sampling valve C. Seven hundred and fifty milliliters of petroleum ether (boiling point 95° to 130°F) shall then be washed through the contaminant mixing chamber and test filter unit using a wash bottle. The wash fluid shall be collected in the same 5,000-ml beaker.

h. A Type AA-047 Millipore membrane filter paper disk shall be washed with 200 ml of petroleum ether and dried to constant weight at 125°F. The weight shall be accurate to 0.1 mg.

i. The weighed filter membrane shall be assembled into the filter holder assembly (see figure 4) and fitted to a 5,000-ml vacuum flask connected to a vacuum pump (minimum 25 inches Hg vacuum).

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Size Range (Microns)	Number (by Count)	Weight (Milligrams)	Percent (by Weight)
0 - 2	76×10^6	0.2	0.04
2 - 5	190×10^6	11.5	2.3
5 - 10	225×10^6	110.5	22.1
10 - 15	101×10^6	187.4	37.4
15 - 20	17×10^6	95.2	19.0
20 - 25	38×10^6	46.3	9.3
25 - 30	120×10^3	2.8	0.6
30 - 40	85×10^3	4.2	0.8
40 - 50	70×10^3	7.5	1.5
50 - 60	50×10^3	9.6	1.9
60 - 70	40×10^3	12.8	2.6
70 - 80	25×10^3	12.0	2.4
		500.0	100.0

Points on the Line Indicate Portions of Particles with Sizes Greater Than Diameter Noted.

FIGURE 6. Particle Size Distribution Curve APM F-9 Glass Beads

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j. All fluid passed through the test filter shall be filtered through the membrane disk. The 5,000-ml beaker shall then be washed with 500 ml of petroleum ether (boiling point 95° to 130°F), and this fluid from the washed beaker shall pass through the membrane. The filter funnel shall be washed with 100 ml of petroleum naphtha and 100 ml of petroleum ether and this wash passed through the membrane.

k. Maintaining vacuum, the top half of filter funnel assembly shall be removed, exposing the filter paper for further washing.

l. With a wash bottle of filtered petroleum ether (boiling point 95° to 130°F), the rim of the filter paper shall be gently washed to remove the traces of oil, being careful not to disturb the contaminant cake. Vacuum shall be maintained during the above operation.

m. The vacuum shall be shut off and the filter paper removed. It shall be dried to a constant weight at 125°F and weighed to an accuracy of a tenth of a milligram. The difference between the filter paper weights before and after the above procedure shall be the weight of contaminant passing through the test filter B in the formula below. The above test, with the element removed, shall be repeated four times for four separate contaminant add values (A in formula below). Oil cleaned to a level that conforms to 4.6.6.2.1 shall be used for each test. The contaminant collected shall be the add value and shall be used in the calculation of degree of efficiency. It shall be the average of the four runs. A system blank value shall be obtained by repeating steps a through m above with the filter element installed and no contaminant added. This blank value shall be the value C in the following efficiency formula and shall be less than 0.0007 gram:

$$\% \text{ removal} = \frac{(A-(B-C)}{A} \times 100$$

No one of the four add values shall be less than 95 percent by weight of the contaminant introduced.

n. The filter unit shall remove 99.3 percent by weight of the contaminant add value. This is equivalent to 100-percent removal of particles over 5 microns and 96.6 percent removal of particles 5 microns and smaller in size, by weight.

4.6.6.3 Largest particle passed test. A filtration test shall be conducted as follows:

a. A degree of filtration test shall be conducted in accordance with the procedure specified in 4.6.6.2.2 a through e with 2,000 ml of hydraulic fluid. The combined added contaminant shall be 1.5 grams of APM F-9 beads and 1.5 grams of carbonyl iron E.

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b. The effluent shall be passed through an absolute 1.0- micron membrane disk Type AA Millipore filter paper, or equal.

c. The entire disk shall be scanned. The largest bead shall be no greater than 5 microns.

4.6.6.4 Pressure buildup and collapse test. A pressure buildup and collapse test shall be conducted as follows:

a. The test fluid used in the system shown on figure 7 shall be precleaned by flowing through a cleanup filter using the test procedure specified in 4.6.6.2.1. The cleanup filter shall be bypassed throughout the remainder of this test.

b. The element under test shall be installed in the housing.

c. The pump shall be started, valves A and B opened, and rated flow attained by adjusting the variable speed drive.

d. Standardized fine air cleaner (AC) test dust or carbonyl iron E shall be added in increments of 4 grams at 4-minute intervals through the dust valve, in slurry. Test dust shall have the following composition:

Size of Particle (Micron)	Percent by Weight of Total
0 to 5	39 ±2
5 to 10	18 ±3
10 to 20	16 ±3
20 to 40	18 ±3
Over 40	9 ±3

e. Two minutes after each dust addition, pressure differential and flow and temperature shall be recorded. Dust shall be added until 40 grams of dust are added. The differential pressure across the filter unit shall not exceed 115 psi. The pump shall not be stopped during the test. Then, 200 grams of carbonyl iron E (1 to 12 microns) and glass beads shall be added in equal proportion until a differential pressure of 5,000 psi is obtained across the element. The dirt capacity to 5,000 psi shall be recorded. The effluent shall be sampled and tested in accordance with 4.6.6.3a and b as the differential reaches 100, 150, 1000, 2000, 3000, and 5000 psi. The samples collected at 100 and 150 psi differential shall show a maximum particle size of 5 microns. The samples collected at 1000-, 2000-, 3000-, and 5000-psi differential shall show a maximum particle size of 18 microns.

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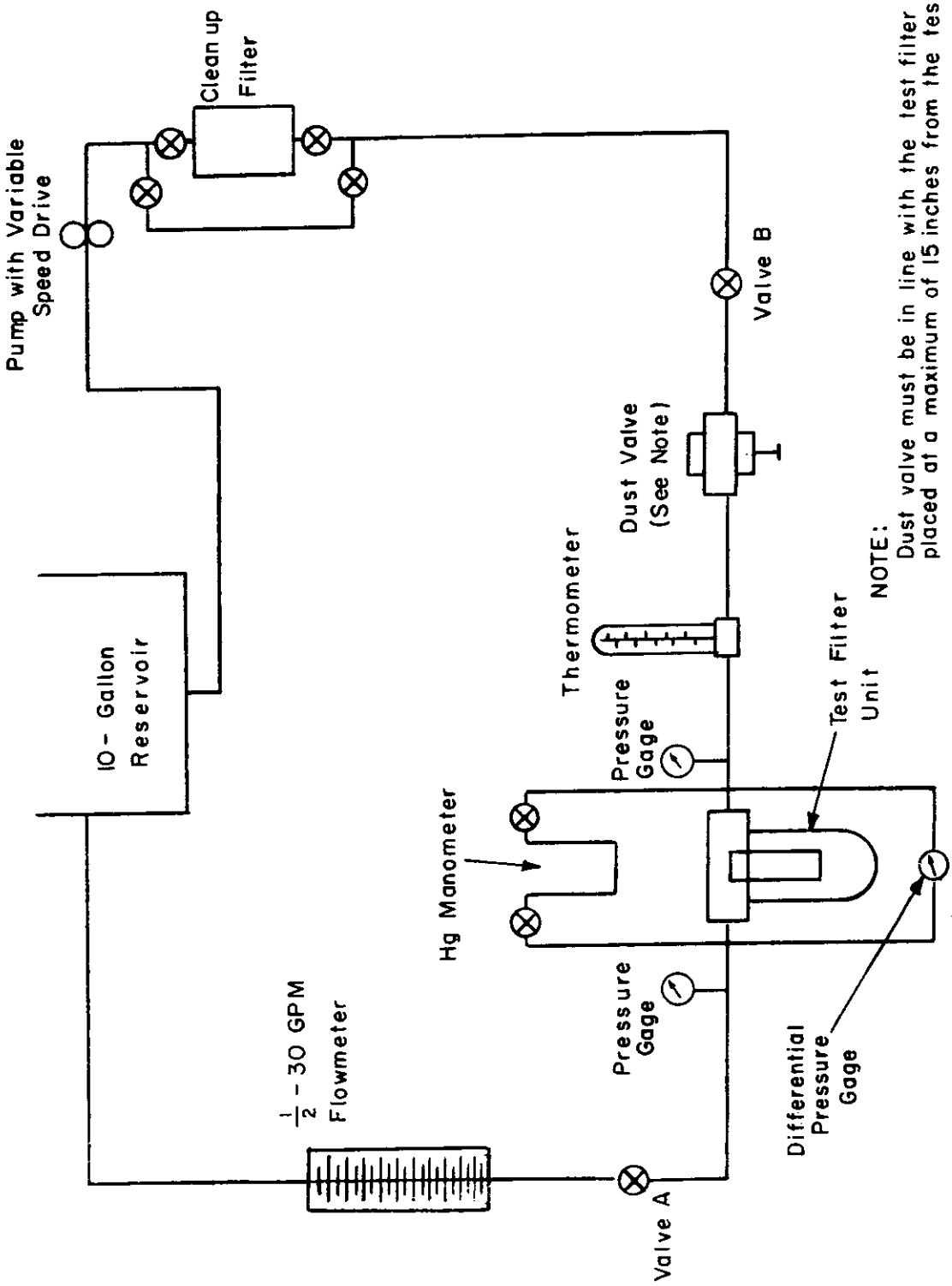


FIGURE 7. Typical Setup for Determining Pressure Buildup and Collapse Pressure Characteristics

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f. Cleanable elements shall be loaded 10 times to their rated differential pressure and cleaned 10 times. After the final cleaning, the dirt-holding capacity shall be not less than 90 percent of the original dirt-holding capacity at the elements rated differential pressure. Elements shall be cleaned by chemicals and ultrasonic equipment in accordance with recommendations submitted by the manufacturer.

g. Cleanable elements shall be subjected to the test specified in 4.6.6.1.

4.6.6.5 Media migration and vibration tests. The filter unit shall be mounted on a test stand and the oil in the test system shall be precleaned using the procedure specified in 4.6.6.2.1. The filter unit, filled to the top of the element with fluid, shall be maintained at 160°F for 72 hours.

4.6.6.5.1 Vibration test. The filter unit shall be cooled to room temperature in three mutually perpendicular planes. The frequency of vibration shall be varied from 50 to 500 Hz, with one pass of scanning to be at least 15 minutes, at an acceleration of 10g in each plane with any resonating frequencies noted. The filter unit shall be vibrated for 1 hour at each of these resonating frequencies. If no resonance is encountered, the vibrating frequency shall be 500 Hz for 1 hour at an acceleration of 10g in each plane. The differential pressure indicator shall not actuate during this test.

4.6.6.5.2 Media migration test. At the completion of the vibration period and with the test pumping system and valves preset to give rated flow through the filter, the system shall be started and 1 gallon of fluid passed through the filter and effluent collected. The sample effluent shall be completely filtered through type AA Millipore paper (47-mm disk) and the amount collected and scanned under a microscope. There shall be no migration traceable to the element or its packaging. After this test, the filter shall successfully withstand the test specified in 4.6.6.1.

4.6.7 Pressure impulse tests

4.6.7.1 Filter housing test. The filter housing shall be subjected to 100,000 impulse cycles at 160°F. Each impulse cycle shall consist of a pressure rise from operating pressure (3,000 psi) to a peak surge pressure of 1.33 $\pm$0.05 times the rated operating pressure, as shown by an oscilloscope. The impulse time (duration of impulse) shall not be longer than 0.05 second, and the rate of pressure rise shall be a minimum of 50,000 psi per second. Cycling shall be performed at a maximum rate of 300 cycles per minute. The differential pressure indicator shall not actuate during the impulse test. There shall be no evidence of external leakage or structural failure during this test.

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4.6.7.2 Filter element test. The filter element impulse test shall be run simultaneously with the housing impulse test. Noncleanable elements shall withstand 10,000 impulse cycles without evidence of structural failure. Cleanable elements shall withstand 100,000 cycles without evidence of structural failure.

4.6.8 Flow fatigue tests

4.6.8.1 Elements to be submitted for flow fatigue tests shall be subjected to the test of 4.6.6.1 to establish an initial bubble point pressure.

4.6.8.2 Filter element test. Filter elements shall be installed in the housing and subjected to 3,000 psi inlet pressure and rated flow with fluid at 160°F. Elements shall be loaded with AC fine test dust and glass beads to produce the specified differential pressure across the housing inlet and outlet. Elements or parts of elements that are noncleanable shall withstand series I only. Elements or parts of elements that are cleanable shall withstand series II only. A cycle shall consist of increasing the differential pressure across the filter unit from 0 to the values specified and back to 0 by first increasing then decreasing flow through the filter unit. The cycling rate shall not exceed 300 cycles per minute. The number of cycles at each differential pressure shall be as follows:

Series I

a. 3,000 cycles at 55 psi pressure drop

b. 6,000 cycles at 75 psi pressure drop

c. 1,000 cycles at 100 psi pressure drop

Series II

a. 40,000 cycles at 75 psi pressure drop

b. 60,000 cycles at 100 psi pressure drop

c. 100 cycles at 3,000 psi pressure drop

4.6.8.2.1 Near the completion of each subseries test a 2,000-ml sample of effluent fluid shall be collected. The fluid sample shall be passed through an absolute 1.0-micron membrane disk type AA Millipore filter paper, or equal. The disk shall be scanned. Any particle larger than the original micron rating of the element under test shall be cause for rejection.

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4.6.8.2.2 At the completion of each subseries test, elements shall be subjected to the test specified in 4.6.6.1. A decrease in bubble pressure from the initial value, as recorded during the test specified in 4.6.8.1, shall be cause for rejection.

4.6.9 Burst pressure test. The burst pressure test shall be conducted at 160°F after a 5-hour soak at 160°F with the filter unit filled with test fluid. Pressure shall be applied with a handpump at a maximum rate of increase of 25,000 psi per minute until the burst pressure of 11,250 psi is reached. Filter units shall show no leakage in the form of drops or rupture of internal or external parts at this pressure when held for 2 minutes. Following this test, the pressure may be increased above that specified in order to secure data on actual burst pressure.

4.6.10 Fluid composition test. The cleanup filter shall be removed from the setup shown on figure 7. The reservoir and circuit shall be filled with unfiltered hydraulic fluid. Two 1-quart samples of the unfiltered fluid shall be collected. The fluid shall then be circulated through the circuit at 30 gpm and two 1-quart samples of the effluent taken. Test dust shall be added at the dust valve until the differential pressure across the filter unit reaches 100 psi. Two 1-quart samples of the effluent shall then be collected. One of each of the three samples of fluid shall be properly identified and shall be submitted to all the tests of MIL-H-5606. The three remaining samples shall be properly identified and forwarded to the procuring activity. Any change to the composition of the hydraulic fluid as a result of passing through the filter shall be cause for rejection.

4.7 Inspection of the preservation, packaging, packing, and marking for shipment and storage. The inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 Packaging (see 6.2)

5.1.1 Level A. One each filter unit or element shall be packaged in accordance with method II of MIL-P-116. All openings of the filter unit or element shall be plugged or sealed with approved noncorrosive materials to prevent the entry of moisture, dirt, and foreign matter. The filter unit or element shall be wrapped in grade A barrier material conforming to MIL-B-121 and placed in a container conforming to PPP-B-636.

5.1.2 Level C. The filter or element shall be packaged in accordance with standard commercial practice.

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5.2 Packing (see 6.2)

5.2.1 Insofar as practical, shipping containers shall be of uniform size and shape and of minimum cube and tare weight. Each shipping container shall contain only one filter unit, or one element when shipping only filter elements.

5.2.2 Level A. Unless otherwise specified, shipping containers shall be nailed wooden or wood-cleated plywood boxes conforming to PPP-B-621 or PPP-B-601 (overseas type).

5.2.3 Level C. Unless otherwise specified, the unit package shall be packed in substantial commercial shipping containers constructed to insure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery. Except as specified herein, the containers shall conform to the requirements of consolidated freight classification rules in effect at the time of shipment.

5.3 Shipment marking. In addition to any special marking required by the contract or order (see 6.2), all shipping containers shall be marked in accordance with MIL-STD-129.

a. The nomenclature for the filter unit shall be as follows:

FILTER UNIT, FLUID, PRESSURE MXU-408/M, ABSOLUTE 5 MICRON HYDRAULIC

b. The nomenclature for the element shall be as follows:

ELEMENT FOR FILTER UNIT, FLUID, PRESSURE MXU-408/M ABSOLUTE 5 MICRON, HYDRAULIC

6. NOTES

6.1 Intended use. The MXU-408/M filter unit covered by this specification is intended for use in the high-pressure outlet line of hydraulic test equipment.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Selection of applicable levels of packaging and packing. (See section 5.)

c. Special shipment marking. (See 5.3.)

6.3 Qualified products list. With respect to products requiring qualification, awards will be made only for products which are at the time set for

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opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division, (Attn: ASNMS-20), Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

6.4 International standardization. This entire specification is the subject of international agreement Air Standardization Co-ordinating Committe Air Standard 11/14. Certain provisions of paragraphs 1.1, 3.7.2.2, 4.6.6.2.2n, and 4.6.6.3c of this specification are the subject of international standardization agreement North Atlantic Treaty Organization Standardization Agreement No. 3510. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

6.5 Abbreviations. In order to save space and to avoid the spelling out of repetitious words and phrases, the following abbreviations have been used in the text of this specification:

ac	alternating current
dc	direct current
dia	diameter
g	gravitational unit
gpm	gallons per minute
Hg	mercury
Hz	Hertz, cycles per second
mg	milligram
min	minimum
ml	milliliter
mm	millimeter

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psi . . . . . pounds per square inch

psig . . . . . pounds per square inch gage

rpm . . . . . revolutions per minute

V . . . . . volt

6.6 Identification of changes. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 11

Review activity:
Air Force - 82

Preparing activity:
Air Force - 11

Project No. 4330-F0065

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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